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SCIENCE PRO

CLASS 10 · PHYSICS · CHAPTER 2

Lenses — PYQs

This is a SAMPLE set of exam-style questions based on the Lenses chapter, written to demonstrate the download feature. They are not copied from an actual question paper. Replace this file with your real previous year question paper via VSCode — keep the filename **lenses-pyqs.pdf** so the site picks it up automatically. Use the Cartesian sign convention and the lens formula 1/f = 1/v - 1/u.

Section A — One-mark questions

1	What is the SI unit of the power of a lens?
2	Name the type of lens that is thicker at the centre than at the edges.
3	What is the focal length of a lens whose power is +2 D?
4	Where must an object be placed so that a convex lens forms an image at infinity?
5	Write the formula for the magnification produced by a lens.

Section B — Two-mark questions

1	Distinguish between a real image and a virtual image.
2	Draw a ray diagram to show the image formed by a convex lens when the object is placed between F and 2F. State the nature, position and size of the image.
3	State the three rays used to locate an image in a convex lens ray diagram.
4	Which type of image is formed by a magnifying glass? State where the object must be placed.
5	The magnification of a lens is -1. What does this tell you about the image?

Section C — Numerical problems

1	An object is placed 25 cm from a convex lens of focal length 20 cm. Find the image distance and the magnification. State the nature of the image.
2	A convex lens has a focal length of 10 cm. An object is placed 30 cm from the lens. Find the position of the image and the magnification.
3	A 4 cm tall object is placed 15 cm in front of a convex lens of focal length 10 cm. Find the position, nature and height of the image.
4	An object is placed 6 cm from a convex lens of focal length 10 cm. Find the image distance, magnification and the nature of the image.
5	Find the power of a convex lens of focal length 25 cm.

Answer key

A1	Dioptre (D)
A2	Convex lens
A3	f = 1/P = 0.5 m = 50 cm
A4	At the principal focus F
A5	$m = h_i/h_o = v/u$
B1	A real image forms on the opposite side and can be caught on a screen; a virtual image forms on the object's side and cannot.
B2	Image beyond 2F, real, inverted and enlarged.
B3	Ray parallel to the axis passes through the far focus; ray through the optic centre goes straight; ray through the near focus emerges parallel to the axis.
B4	Virtual, erect and enlarged. Object between F and the lens.
B5	The image is real, inverted and the same size as the object.
C1	1/v = 1/f + 1/u = 1/20 - 1/25, so v = +100 cm. m = v/u = -4. Real, inverted, enlarged.
C2	1/v = 1/10 - 1/30, so v = +15 cm. m = -0.5. Real, inverted, diminished.
C3	1/v = 1/10 - 1/15, so v = +30 cm. m = -2, so h_i = -8 cm. Real, inverted, 8 cm tall.
C4	1/v = 1/10 - 1/6, so v = -15 cm. m = +2.5. Virtual, erect, enlarged.
C5	P = 1/f(m) = 1/0.25 = +4 D